

3.4.2 Input and Output Handling

print () اسکرین پر معلومات دکھاتا ہے اور input () صارف سے معلومات لیتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 09 of 24

3.4.2 Input and Output Handling

LEARN IT

Output — the `print()` function. A **function** is a named, ready-made piece of code that does a job when you call it with brackets. `print()` displays information on the screen.

- Items separated by commas are printed with one space between them.
- An **f-string** (a string with `f` before the quote) lets you put variables directly inside `{ }`. It is the fastest and most readable way to format output.

Input — the `input()` function. It shows a message (a **prompt**), **pauses the program** and waits for the user to type something and press Enter. Whatever the user types is **always a string** by default.

Type casting means converting a value from one type to another: `int()` converts to an integer, `float()` to a decimal number, `str()` to a string.

```
# 1. Print only a string
print("Hello Class!")

# 2. Print a string and a variable (comma separates them)
name = "Ali"
print("Welcome,", name)

# 3. f-strings (fastest and most readable way to format)
age = 16
print(f"My name is {name} and I am {age} years old.")
```

OUTPUT

```
Hello Class!
Welcome, Ali
My name is Ali and I am 16 years old.
```

Two useful extras: `sep` changes the separator between items, and `end` changes what is printed at the end (normally a new line).

```
print("10", "09", "2026", sep="-")
print("Loading", end="...")
print("done")
print(f"Total: {7 * 3}")
```

OUTPUT

```
10-09-2026
Loading...done
Total: 21
```

WORKED EXAMPLE

Taking input with type casting.

```
# Taking text input
user_name = input("Enter your name: ")
# Type casting: converting string input to an integer (int)
age = int(input("Enter your age: "))
# Type casting: converting string input to a decimal (float)
height = float(input("Enter your height: "))
print(f"Hello {user_name}! Next year you will be {age + 1}.")
print("Height:", height, "feet")
```

Sample run (the user types Sara, 16 and 5.4):

```
OUTPUT
Enter your name: Sara
Enter your age: 16
Enter your height: 5.4
Hello Sara! Next year you will be 17.
Height: 5.4 feet
```

Why type casting matters. Without `int()`, the numbers stay as text and `+` joins them:

```
a = input("First number: ")
b = input("Second number: ")
print(a + b)
print(int(a) + int(b))
```

OUTPUT

```
First number: 5
Second number: 3
53
8
```

"5" + "3" joins two strings into "53" (called **concatenation**). `5 + 3` adds numbers to give 8.

REMEMBER IT

"input() always gives a string." If you plan to do maths, wrap it: `int(input(...))` for whole numbers, `float(input(...))` for decimals. Read it from the inside out: first `input`, then convert.

COMMON MISTAKE

If the user types letters where a number is expected, `int()` crashes with `ValueError: invalid literal for int() with base 10` (see 3.7). Also, `int("5.4")` fails — use `float()` for decimal input.

اردو میں سمجھیں

`print()` اسکرین پر معلومات دکھاتا ہے اور `input()` صارف سے معلومات لیتا ہے۔ یاد رکھیں کہ `input()` ہمیشہ اسٹرنگ (String) دیتا ہے، اس لیے حساب کے لیے اسے `int()` یا `float()` سے تبدیل کرنا ضروری ہے۔ اسے ٹائپ کاسٹنگ (Type Casting) کہتے ہیں۔ ایف اسٹرنگ (f-string) سے ویری ایبلز کو جملے کے اندر آسانی سے لکھا جا سکتا ہے۔

CHECK YOURSELF

1. Predict the output:

```
a = "3"
b = 4
print(a * b)
print(int(a) * b)
```

2. Find the error: the user types 21 but the output is 2121, not 42.

```
num = input("Enter a number: ")
print(num * 2)
```

Answers: 1. 3333 then 12 (a string times a number repeats the string). 2. `input()` returns the string "21", and a string times 2 repeats it. Fix: `num = int(input("Enter a number: "))`.

Next lecture: 3.4.3 Operators and Operands: Arithmetic and Assignment